

Trust-Calibrated Autonomy for Operational AI Agents in Production Environments. A Design Science Framework and Practitioner Validation Protocol

Structured abstract

Context. Production operations are becoming more complex as organizations rely on distributed systems, platform engineering, and observability stacks that require high-tempo interpretation of logs, metrics, traces, and service dependencies (Beyer et al., 2016, 2018; Sridharan, 2018; Skelton & Pais, 2019; DORA, 2024). At the same time, AIOps is moving from anomaly detection and alert correlation toward large language model, LLM, based agents that can summarize telemetry, generate hypotheses, and support root cause analysis and incident handling (Dang et al., 2019; Notaro et al., 2021; Ahmed et al., 2023; Chen et al., 2024; Roy et al., 2024; Zhang et al., 2024; Zhang et al., 2025).

Objective. This paper addresses a socio-technical question that remains underdeveloped in current AIOps literature: how should organizations calibrate appropriate autonomy levels for AI agents operating in production environments, where action quality, reversibility, and blast radius matter as much as model capability (Lee & See, 2004; Hoff & Bashir, 2015; Glikson & Woolley, 2020; Steinmetz et al., 2025; Regona et al., 2026)?

Method. The study follows design science research and develops a reusable artifact, the Operational Trust Calibration Framework, through structured literature synthesis across software operations, observability, AIOps, trust in automation, human-AI interaction, and AI governance (Hevner et al., 2004; Peffers et al., 2007; Gregor & Hevner, 2013; Wieringa, 2014). The review protocol is informed by established software engineering review guidance and PRISMA 2020 (Kitchenham, 2004; Kitchenham & Charters, 2007; Brereton et al., 2007; Petersen et al., 2015; Wohlin, 2014; Wohlin et al., 2020; Page et al., 2021). Because practitioner interviews are not yet completed, the validation component is presented as a structured protocol rather than as completed empirical results.

Results. The paper contributes a vendor-neutral framework that combines three operational constraints, **observability, reversibility, and blast radius**, with a governance envelope consisting of approval authority, auditability, override capability, policy fit, evidence requirements, and staged deployment rules. The artifact defines four autonomy levels, from summarize-only support to conditional bounded autonomy, and formulates design principles for safe autonomy escalation in production environments.

Conclusion. Trust in operational AI should not be modeled as a general attitude toward intelligent tools. In production operations, trust must be calibrated as a bounded organizational decision about what an AI agent is allowed to know, recommend, and do under specific technical and governance conditions. The proposed framework offers a theoretically grounded and operationally actionable basis for that calibration.

Keywords: trust calibration, AIOps, platform engineering, observability, incident response, operational AI, design science research, AI governance

1. Introduction

Modern production environments are difficult to understand and even harder to govern under incident conditions. Distributed architectures, cloud platforms, Kubernetes clusters, CI/CD pipelines, internal developer platforms, and polyglot telemetry pipelines have increased both observability potential and operational cognitive load (Beyer et al., 2016, 2018; Sridharan, 2018; Skelton & Pais, 2019). Platform engineering aims to reduce this complexity through standardization, paved roads, and product-oriented internal platforms, yet the latest DevOps evidence suggests that operational complexity still outpaces many organizations’ ability to manage it consistently (DORA, 2024). OpenTelemetry and related semantic conventions have improved telemetry standardization, but more data does not automatically yield better incident decisions (OpenTelemetry Authors, n.d.-a, n.d.-b; Sridharan, 2018).

In parallel, AIOps has undergone a marked transition. Earlier work focused on alert correlation, anomaly detection, and isolated failure-management tasks (Dang et al., 2019; Notaro et al., 2021). More recent work shows that LLM-based systems can support or automate parts of incident diagnosis and mitigation, including root-cause recommendation, RCA narrative generation, tool-augmented reasoning, and agentic problem solving (Ahmed et al., 2023; Chen et al., 2024; Roy et al., 2024; Yao et al., 2022; Zhang et al., 2024). The latest survey literature confirms rapid growth in LLM-centered AIOps and notes that root cause analysis, log analysis, and incident triage dominate the field’s recent attention (Zhang et al., 2025). Benchmark work such as AIOpsLab likewise shows that the research frontier is moving toward end-to-end evaluation of autonomous operational agents rather than isolated prediction tasks (Chen et al., 2025). Recent analyses of agentic AI frameworks confirm that most current frameworks remain immature, with limited memory, basic planning capabilities, and incomplete enterprise features such as observability and monitoring (Vaidhyanathan & Taibi, 2026). Systematic reviews of AIOps for log anomaly detection similarly note that despite strong LLM capabilities, data confidentiality concerns and multi-task adaptiveness remain significant barriers to adoption (De la Cruz Cabello et al., 2025).

Yet an important gap remains. Most current work asks whether AI agents can diagnose incidents, retrieve relevant evidence, or recommend remediation. Much less work asks how an organization should decide **whether those agents should be allowed to act**, under what constraints, and with what oversight. This is the central challenge in production settings. A wrong recommendation may cost attention and time. A wrong autonomous action may expand an outage, erase evidence, violate control boundaries, or increase public harm. In other words, the key problem is not only model capability. It is **autonomy calibration**.

Trust research provides essential foundations for this problem. Organizational trust has long been described as willingness to accept vulnerability based on expectations of another party’s ability, benevolence, and integrity (Mayer et al., 1995; McKnight et al., 1998; Schoorman et al., 2007). Trust in automation research shifted that logic into socio-technical settings, showing that misuse and disuse emerge when reliance does not match actual system capability (Lee & Moray, 1992; Muir, 1994; Muir & Moray, 1996; Parasuraman et al., 2000; Jian et al., 2000; Lee & See, 2004; Merritt & Ilgen, 2008; Hoff & Bashir, 2015). More recent trust-in-AI work extends these themes into organizational and human-AI contexts, emphasizing that trust must be made legible

through performance characterization, transparency, usability, and safety rather than treated as an unexamined psychological residue (Glikson & Woolley, 2020; Gillath et al., 2021; Lukyanenko et al., 2022; Steinmetz et al., 2025; Regona et al., 2026).

At the same time, human-AI interaction work shows that explanations alone do not guarantee appropriate reliance. Explanation quality, behavioral descriptions, calibration cues, and interface design all shape whether users under-rely on or over-rely on AI advice (Ribeiro et al., 2016; Doshi-Velez & Kim, 2017; Gunning & Aha, 2019; Amershi et al., 2019; Bansal et al., 2021; Göbel et al., 2022; Schemmer et al., 2023; Vasconcelos et al., 2023; Cabrera et al., 2023; Miller, 2019). These insights are reinforced by current software engineering research on AI-powered developer tools, which shows that trust shapes adoption, validation effort, willingness to delegate, and even how developers renegotiate their professional roles around AI assistance (Johnson et al., 2023; Wang et al., 2024; Murillo et al., 2024; de Souza et al., 2024; Basha & Rodríguez-Pérez, 2025; Kemell et al., 2025; Sabouri et al., 2025).

However, production operations differ from many other human-AI settings. Incidents unfold under time pressure, uncertainty, and cross-team dependencies. They also take place in settings that resemble high-reliability organizing, where coordination, heedful interrelating, and accountability are central to resilient action (Weick & Roberts, 1993). Therefore, trust calibration for operational AI cannot be reduced to interface niceties or generic model transparency. It must be translated into an organizational decision model for granting, constraining, and auditing autonomy in live systems.

This paper addresses that problem by asking: **What trust and governance mechanisms enable organizations to calibrate appropriate autonomy levels for AI agents operating in production environments?** The paper contributes a design-science artifact, the **Operational Trust Calibration Framework**, that links operational constraints to staged autonomy rights and governance controls. The framework will be validated through structured expert interviews at a government-sector infrastructure organization in Estonia where the first author serves as Chief Infrastructure Architect, providing direct access to the operational environments, incident workflows, and practitioner perspectives needed for empirically grounded evaluation.

2. Theoretical background

2.1 Trust calibration, autonomy, and appropriate reliance

Trust research offers three important lessons for operational AI. First, trust is relational and contextual, not absolute (Mayer et al., 1995; McKnight et al., 1998). Second, automation research shows that reliance should be calibrated to actual capability rather than maximized in the abstract — miscalibration leads to misuse (over-reliance) or disuse (underuse), both of which harm outcomes (Lee & See, 2004; Hoff & Bashir, 2015). Third, levels of automation matter because they redistribute cognitive work, authority, and responsibility across human and machine actors (Parasuraman et al., 2000).

These findings are especially relevant in operations. A recommendation system used during incident triage does not pose the same risk profile as an agent authorized to restart services, alter traffic routing, or trigger rollback procedures. The latter

changes not only human workload but also accountability, evidentiary traceability, and organizational exposure. This makes autonomy a governance issue as well as a human-factors issue.

Recent trust-in-AI research extends these foundations, showing that trust increasingly depends on how capabilities, limitations, and operating conditions are communicated, not merely on whether performance metrics exist (Glikson & Woolley, 2020; Lukyanenko et al., 2022; Regona et al., 2026; Steinmetz et al., 2025). Human-AI interaction work demonstrates that appropriate reliance depends on explanations that are actually usable and interfaces that help users judge when to verify, when to comply, and when to override (Amershi et al., 2019; Bansal et al., 2021; Schemmer et al., 2023; Vasconcelos et al., 2023; Cabrera et al., 2023). For operational AI, this means that model confidence alone is insufficient — production use requires telemetry-grounded explanations, known execution boundaries, and visible control mechanisms. A trustworthy incident agent is not merely accurate; it is also governable.

2.2 Governance and trustworthy AI in operational settings

Current governance frameworks reinforce this point. The NIST AI RMF positions trustworthy AI as a socio-technical objective supported by governance, mapping, measurement, and management activities (NIST, 2023). The Generative AI Profile adapts those concerns to generative systems, where hallucination, opaque generation, information leakage, and misuse risks become more salient (NIST, 2024a). CSF 2.0 and the revised incident response guidance further emphasize that incident response capability should be tightly integrated with risk management, control accountability, and recovery planning (NIST, 2024b, 2025). ISO 31000 and ISO/IEC 42001 similarly frame risk and AI governance as organizational systems rather than isolated technical mechanisms (ISO, 2018; ISO/IEC, 2023).

The governance implication is straightforward. Production AI agents should not be granted autonomy because they appear useful. They should be granted bounded autonomy only when their action space, failure modes, and control envelope are sufficiently understood.

3. Research method

This study follows **design science research**, because its main output is an artifact intended to solve a practically significant organizational problem while remaining theoretically grounded (Hevner et al., 2004; Peffers et al., 2007; Gregor & Hevner, 2013; Wieringa, 2014). The artifact in this paper is the Operational Trust Calibration Framework. Its purpose is to help organizations determine appropriate autonomy levels for operational AI agents in production environments.

The knowledge base for the artifact was assembled through a **structured literature review** that integrates four streams:

- (1) software operations, observability, and platform engineering;
- (2) AIOps and LLM-based incident-management systems;
- (3) trust in automation, trust in AI, and appropriate reliance; and
- (4) AI governance, cybersecurity risk management, and management system standards.

The review design follows established software engineering review guidance and

PRISMA 2020 reporting logic (Kitchenham, 2004; Kitchenham & Charters, 2007; Brereton et al., 2007; Petersen et al., 2015; Wohlin, 2014; Wohlin et al., 2020; Page et al., 2021). Search and screening were conceptually oriented toward combinations of terms such as *AI Ops*, *operational AI*, *incident response*, *root cause analysis*, *trust in automation*, *trust in AI*, *appropriate reliance*, *governance*, *risk management*, and *design science*. The goal of the review was not bibliometric exhaustiveness alone, but requirements extraction for artifact design.

Because practitioner interviews are not yet complete, this paper does **not** claim completed validation findings. Instead, it contributes a **validation protocol** that specifies how the artifact should be stress-tested through structured expert interviews and incident-based constraint analysis in the next stage of the research.

4. Artifact: The Operational Trust Calibration Framework

The core proposition of this paper is that operational AI trust should be calibrated through a bounded decision model, not through generic confidence in AI. The proposed artifact therefore combines three operational constraints with a governance envelope.

4.1 Core calibration dimensions

The first dimension is **observability**. Observability refers to the degree to which the agent and its human supervisors can ground their reasoning in sufficiently rich and relevant telemetry, including logs, metrics, traces, dependency context, recent changes, and service metadata (Sridharan, 2018; OpenTelemetry Authors, n.d.-a, n.d.-b). High observability improves the inspectability of recommendations and supports post-hoc accountability.

The second dimension is **reversibility**. Reversibility refers to how safely, quickly, and completely an action can be rolled back. This includes technical rollback, state recovery, and the ability to restore a prior stable configuration. Reversibility matters because even a highly probable action should not be delegated if its error cost is effectively irreversible (Lee & See, 2004; Hoff & Bashir, 2015).

The third dimension is **blast radius**. Blast radius refers to the likely operational, organizational, legal, and public consequences if the action is wrong. In production environments, blast radius is multi-layered. It includes system impact, customer impact, cross-service dependency effects, evidentiary loss, compliance consequences, and reputational damage (NIST, 2024b, 2025; ISO, 2018).

These three dimensions form the **ORB kernel**, Observability, Reversibility, Blast Radius. The core calibration logic can be expressed as a conceptual model:

$$\text{Appropriate Autonomy} \propto (\text{Observability} \times \text{Reversibility}) / \text{Blast Radius}$$

As observability and reversibility increase, a higher autonomy level becomes justifiable. As blast radius increases, the acceptable autonomy ceiling drops. This is not a numerical formula but a structured decision heuristic: for each agent-task-context configuration, the ORB assessment determines which autonomy level (Section 4.3) is permissible under the governance envelope (Section 4.2). Figure 1 illustrates the overall framework structure.

Operational Trust Calibration Framework

Mayer et al. (1995) · Lee & See (2004) · Parasuraman et al. (2000)

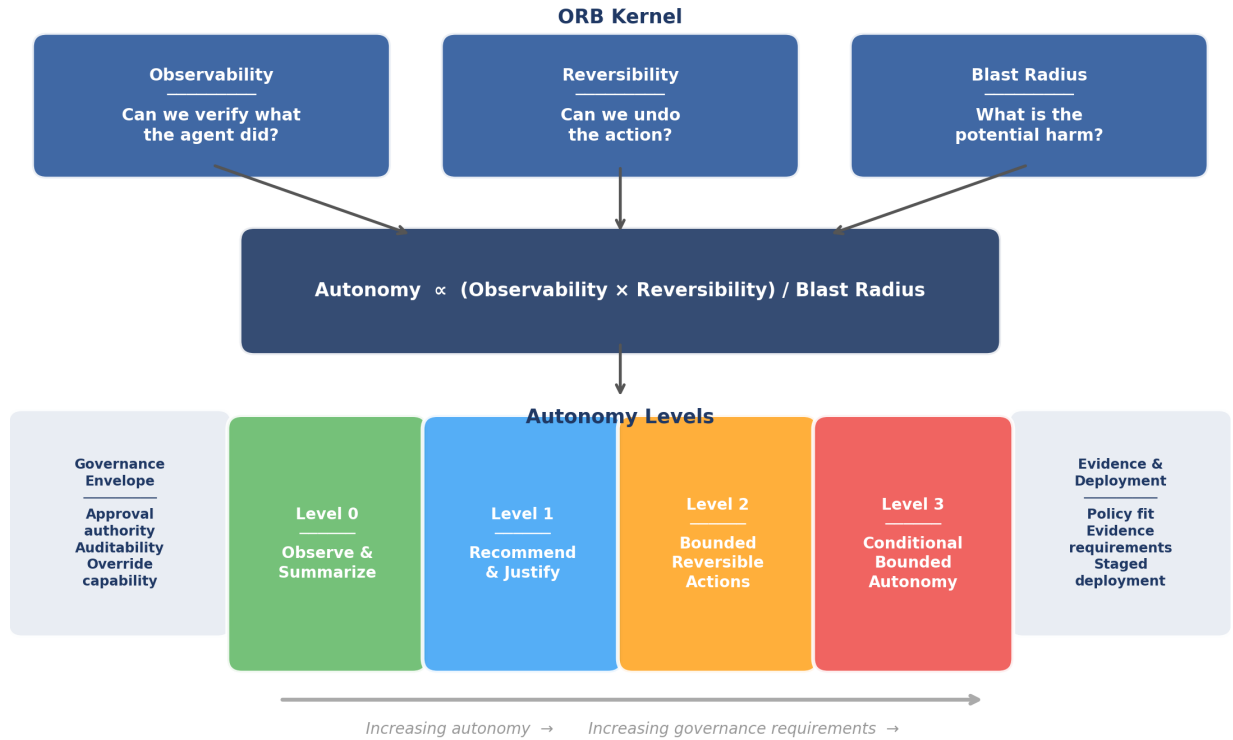


Fig. 1. The Operational Trust Calibration Framework. Three operational constraints (ORB kernel) are assessed against a governance envelope to determine the appropriate autonomy level.

4.2 Governance envelope

The ORB kernel alone is insufficient. It must be wrapped in a governance envelope that defines the conditions under which a given autonomy level is permissible. This envelope includes at least six elements.

First, **approval authority** must be explicit. The framework must specify who may authorize which class of actions and under what incident conditions (NIST, 2023, 2025).

Second, **auditability and provenance** must be preserved. Recommendations, evidence sources, prompts, tool calls, approvals, and actions should be logged in a way that supports review and accountability (ISO/IEC, 2023; NIST, 2023).

Third, **override and kill-switch capability** must be available. Human actors must be able to stop, reverse, or suspend an agent's action pathway when uncertainty increases or context changes (Shneiderman, 2022).

Fourth, **policy fit** must be established. Some incident types, such as security events, cross-organizational outages, or stateful transaction anomalies, may be categorically excluded from automated action even when the agent appears technically competent (NIST, 2024a, 2025).

Fifth, **evidence requirements** must be defined. An agent should not act only be-

cause it produces a plausible narrative. It should meet an explicit evidence threshold tied to telemetry quality, runbook fit, and operational context. Where retrieval-augmented generation (RAG) is used to ground agent reasoning in operational knowledge, the provenance and freshness of retrieved context must themselves be auditable (Lewis et al., 2020; Ribeiro et al., 2016; Amershi et al., 2019; Steinmetz et al., 2025).

Sixth, **deployment staging** must be enforced. Operational AI should progress through offline analysis, replay, shadow mode, recommendation mode, bounded execution with authorization, and only then, where justified, conditional bounded autonomy (Hevner et al., 2004; Peffers et al., 2007; NIST, 2023).

4.3 Autonomy levels

The framework defines four autonomy levels.

Level 0, Observe and summarize.

The agent may retrieve, correlate, summarize, and explain incident evidence, but it may not recommend or execute actions. This level is appropriate when observability is partial, blast radius is high, or governance is not yet mature.

Level 1, Recommend and justify.

The agent may suggest hypotheses, rank likely causes, and propose runbook steps with evidence. Humans remain solely responsible for action selection. This level is suitable when telemetry is informative but uncertainty remains material.

Level 2, Execute bounded reversible actions with authorization.

The agent may carry out predefined, reversible tasks such as restarting a failed worker, scaling a non-critical service, or applying a known-safe rollback, but only after explicit approval by an authorized human.

Level 3, Conditional bounded autonomy.

The agent may act within a pre-approved envelope without case-by-case approval, but only for actions with strong observability, high reversibility, low blast radius, and complete logging and review controls.

This article deliberately excludes broad unsupervised autonomy in production. In regulated or public-impact environments, the burden of justification for such autonomy is far higher than current literature supports (NIST, 2023, 2024a, 2025; ISO/IEC, 2023).

Table 1. Summary of autonomy levels and their ORB conditions.

Autonomy level	Observability	Reversibility	Blast radius	Governance requirement
Level 0: Observe and summarize	Any	N/A (read-only)	Any	Logging only
Level 1: Recommend and justify	Moderate to high	N/A (no action)	Any	Human selects action
Level 2: Execute bounded reversible actions	High	High	Low to moderate	Explicit human approval per action
Level 3: Conditional bounded autonomy	High	High	Low	Pre-approved policy envelope, full audit trail

4.4 Design principles

Five design principles emerge from the synthesis.

DP1. Autonomy should be task-specific, not model-specific.

An AI agent should not receive one global autonomy label. Autonomy should be granted per task class under explicit contextual assumptions (Parasuraman et al., 2000; Lee & See, 2004).

DP2. Trustworthy operational support must be telemetry-grounded.

Natural-language fluency without evidentiary anchoring is dangerous in incidents. Recommendations should be traceable to logs, traces, metrics, topology, or verified operational knowledge. Where retrieval-augmented generation is used, the provenance and freshness of retrieved context must be verifiable (Sridharan, 2018; Ahmed et al., 2023; Chen et al., 2024; Lewis et al., 2020).

DP3. Reversibility should act as a gate, not a bonus.

Where rollback is weak or side effects are durable, autonomy should remain low even if predicted accuracy appears high (Hoff & Bashir, 2015; NIST, 2025).

DP4. High blast radius requires governance escalation.

As impact grows, the acceptable autonomy ceiling should drop and human approval, multi-party authorization, or categorical exclusion should increase (ISO, 2018; NIST, 2024b, 2025).

DP5. Appropriate reliance must be designed, not assumed.

Operational trust depends on explanations, interface cues, behavioral descriptions, and workflow fit that support verification and override rather than passive acceptance (Amershi et al., 2019; Bansal et al., 2021; Schemmer et al., 2023; Vasconcelos et al., 2023; Cabrera et al., 2023).

5. Practitioner validation protocol

The next empirical stage should validate the framework through structured expert interviews and incident-based constraint stress-testing.

5.1 Participants

A maximum-variation expert sample should include incident commanders, SREs, platform engineers, security operations specialists, service owners, and risk or audit representatives. The primary validation site is a government-sector infrastructure organization (SMIT, Estonia) that manages critical digital service infrastructure for public institutions, providing access to operational environments where the framework’s assumptions can be tested against real incident workflows and governance constraints. This composition reflects the fact that operational autonomy is not only a technical decision. It is also a governance and accountability decision (NIST, 2023; ISO/IEC, 2023).

5.2 Procedure

Each session should last approximately 60 to 75 minutes and proceed in four phases:

1. **Role and context elicitation**, focused on current incident workflows, approval boundaries, and existing AI use.
2. **Framework introduction**, where participants review the ORB dimensions and autonomy levels.
3. **Incident vignette stress-testing**, where participants assess realistic scenarios and identify the highest acceptable autonomy level.
4. **Reflective critique**, where they identify missing variables, unacceptable assumptions, and likely adoption barriers.

5.3 Stress-test materials

The vignettes should systematically vary observability, reversibility, and blast radius. Examples include a reversible low-risk worker restart, a customer-facing degradation with incomplete telemetry, a third-party dependency failure, and a security-sensitive incident where evidence integrity is uncertain. For each vignette, participants should rate ORB conditions, choose an acceptable autonomy ceiling, and specify required controls.

5.4 Analysis

Interview analysis should combine matrix comparison with thematic coding. The main analytic outputs should be:

- (a) recurrent autonomy ceilings;
- (b) controls that participants view as non-negotiable;
- (c) role-based disagreements; and
- (d) proposed refinements to the artifact.

6. Interview guide piloting through design probes

Qualitative research methodology recommends piloting interview instruments before field deployment to identify ambiguous questions, missing dimensions, and likely disagreement patterns (Yin, 2018). Because the validation interviews require ethics committee approval before participant recruitment, this study uses fictional personas as **design probes** to stress-test the interview guide and framework logic prior to real data collection.

Six personas were constructed to represent distinct organizational perspectives on operational autonomy: an incident commander (pressuring escalation logic and override rights), a platform product owner (pressuring standardization and policy-as-code), a security operations lead (pressuring exclusion zones, provenance, and data boundaries), a critical service owner (pressuring reputational and legal blast radius), a risk officer (pressuring auditability and documented authority), and an on-call engineer (pressuring evidence quality and workflow fit).

Each persona was used to simulate interview responses and identify which framework dimensions provoke the strongest reactions, which ORB assessments produce disagreement across roles, and where the governance envelope requires additional specificity. These probes are explicitly **not empirical evidence** and are not presented as findings. Their purpose is methodological: to ensure that the interview protocol, when deployed with real practitioners, is capable of surfacing the tensions the framework is designed to resolve.

7. Discussion

The main theoretical contribution of this paper is to move trust calibration from a mostly human-factors problem into a software operations and governance problem. Prior trust-in-automation research explains why appropriate reliance matters, but it does not directly specify how to assign operational action rights in live incident contexts (Lee & See, 2004; Hoff & Bashir, 2015). By contrast, the proposed artifact turns trust calibration into a structured autonomy decision grounded in operational constraints and governance mechanisms.

The paper also contributes to software engineering. Current research on AI-powered developer tools shows that trust shapes uptake, oversight behavior, and validation effort (Johnson et al., 2023; Wang et al., 2024; Murillo et al., 2024; Sabouri et al., 2025). This paper extends that line of thinking from code generation and developer tooling into platform operations, where the consequences of miscalibration can be more immediate and systemic.

The practical contribution is a reusable decision model for organizations that want to move beyond vague “human in the loop” language. The framework specifies **what kind of human control, for which actions, under which evidentiary conditions, and with what reviewability**. This is especially relevant for public-sector and regulated settings, where the social blast radius of operational mistakes may be larger than the purely technical blast radius.

Finally, the paper suggests that the relevant unit of trust is not the agent as such, but the **agent-task-context-control configuration**. This reframing helps avoid a common failure mode in AI adoption, namely granting too much or too little trust based on generic impressions of the tool rather than on bounded operational analysis.

An important limitation of the current framework is that it presents trust calibration as a static assessment. In practice, trust is dynamic: it forms through early interactions, calibrates through demonstrated competence, and can decay or repair after failures (Lee & See, 2004; Hoff & Bashir, 2015). The ORB assessment for a given agent-task configuration should therefore evolve over time as operational evidence accumulates. An agent that consistently produces accurate diagnoses under well-

observed conditions may earn escalation from Level 1 to Level 2; one that causes an incident regression should be demoted. This temporal dimension — how organizations should update autonomy levels based on longitudinal evidence — is beyond the scope of this paper but will be addressed through empirical deployment studies in subsequent work.

8. Limitations

This paper has three clear limitations. First, it does not yet report completed practitioner validation. Second, the framework is vendor-neutral by design, which supports transferability but leaves implementation details for later work. Third, the design probes used before interviews are fictional and should be treated only as rehearsal instruments. These limitations are acceptable at the current stage because the contribution is a design-science artifact plus validation protocol, not a completed field evaluation.

9. Conclusion

Operational AI in production should not be framed as a binary choice between assistance and autonomy. The meaningful question is how much autonomy is justified for which tasks, under which conditions, and with what safeguards. This paper proposes an answer through the Operational Trust Calibration Framework, which combines observability, reversibility, and blast radius with a governance envelope of approval authority, auditability, override, policy fit, evidence requirements, and staged deployment. The framework offers a practical basis for organizations seeking to adopt operational AI without confusing technical capability with organizational readiness.

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